

RAG-READY DOCUMENT

Project: L'ORÉAL

Feature: ZPA Replenishment Process Redesign (Volume-Based Logic)

Version: 2.2

Initial Date: 12/11/2025

Last Update: 14/01/2026

Authors: ZBA / CDE

◆ DOCUMENT METADATA (Vector DB Ready)

```
{  
  "document_type": "Technical & Functional Specification",  
  "project": "L'OREAL",  
  "system": "Speed WMS",  
  "process": "ZPA Replenishment",  
  "feature": "Volume-Based Replenishment and Stock Splitting",  
  "related_modules": ["ZPA", "Replenishment Module"],  
  "entry_point_modified": "PE482",  
  "status_handling": ["01", "03"],  
  "version": "2.2"  
}
```

◆ SECTION 1 — Business Context

1.1 Background

As part of the L'Oréal project, the ZPA replenishment process is being redesigned to:

- Improve handling of partial replenishments
- Consider real volume constraints of storage locations
- Optimize flow between pallets/boxes and dynamic picking locations

1.2 Current Limitations

The existing ZPA process:

- Transfers full supports only
- Does not calculate actual available volume
- Does not split intelligently based on capacity

This causes:

- Lack of shelf space
 - Large variability in article volume (lipsticks vs bottles)
 - No automatic calculation of remaining capacity
 - Manual management of partial replenishments
-

◆ SECTION 2 — Functional Objectives

The workshop defined the following goals:

1. Automatically split replenishments according to destination volume.
 2. Fill picking locations based on volume, not only quantity.
 3. Improve ergonomics in the replenishment module.
 4. Make ZPA replenishment more dynamic (triggered at wave launch).
 5. Manage saturated locations (rejection or deferred handling).
 6. Handle residual quantities via rejection location.
 7. Avoid intrusive changes in standard launch module.
-

◆ SECTION 3 — ZPA Functional Solutions

3.1 New “Free Location” Search Rule

A new free-location search rule will replace the current ZPA logic.

◆ Stock Splitting Logic

If required quantity < support quantity:

SPLIT original_stock_line

NEW_LINE.quantity = replenishment_need

ORIGINAL_LINE remains in original location

◆ Location Eligibility Criteria

For the new stock line, search empty eligible locations based on:

- Destination Zone
 - Original Support Format
 - Quality Codes (include/exclude)
 - Aerosol Codes (ART_ALPHA3 include/exclude)
 - Family Codes (ART_ALPHA11 include/exclude)
 - ART_ALPHA8 include/exclude
 - ART_ALPHA9 include/exclude
-

◆ Threshold Rule

If threshold reached (ART_NUM2):

NO available location returned

◆ Volume Check

Before assigning location:

required_volume = article_volume × quantity_to_store

Article volume must be taken from:

art_dim table – unit dimensions (500)

If required volume fits → assign location.

◆ If Volume Does NOT Fit

Split stock again, keeping:

- Full parcels only
- Limited to available location volume

Parcel volume:

parcel_volume = article_volume × parcel_packaging_quantity

Continue searching other locations until:

- Stock exhausted
OR
 - No more locations available
-

◆ Foisonnement Coefficient

A configurable coefficient at activity level will:

adjusted_volume = calculated_volume × foisonnement_coefficient

This applies to:

- Unit volume
 - Parcel volume
-

◆ SECTION 4 — ZPA Residual (Rejection) Management

If stock cannot be stored:

→ Automatically sent to a destination location of type:

REJET (Rejection)

Behavior Rules

- Stock lines without valid location remain in status 03.
 - They are NOT converted into replenishment mission lines.
-

Entry Point Modification

PE482 (end of ZPA process) must be modified:

GENERATE replenishment missions

ONLY IF destination_location_type ≠ REJET

REJET Lines

Lines linked to REJET location:

- **Remain in status 03**
 - **No mission generation**
-

Contextual Menu

A specific contextual menu will allow operators to:

Change status from 03 → 01

Clear STK_LIEUP

Relaunch ZPA using standard process

◆ SECTION 5 — Replenishment Module Improvements

5.1 Grouping Replenishments by Origin Location

Goal:

Allow operators to process multiple replenishments from the same source location simultaneously.

Sorting Adaptation

Modify radio sorting to group replenishments by:

- **Zone**
 - **Origin location**
-

5.2 Relay Location (Temporary Support)

Operators may:

1. **Scan a relay cart or pallet.**

2. Load parcels onto relay support.
3. Move to picking locations.
4. Scan relay support again for storage.

◆ **SECTION 6 — Operational Feedback (Workshop 21/01/2026)**

◆ **Case: 48 Parcels on a 50-Parcels Pallet**

If split into 20 replenishments:

Problem:

Operator must:

- Scan source location 20 times
- Validate quantity 20 times

Optimization needed to reduce repetitive validation.

◆ **Volume Completion Rule**

Discussion needed on whether to maintain strict volume completion logic.

◆ **Volume Source Clarification**

Confirmed by RBA (21/01/2026):

Article volume must be taken from:

art_dim table – unit dimension (code 500)

◆ **SECTION 7 — Technical Logic Overview**

FOR each ZPA replenishment:

IF need < stock_quantity:

SPLIT stock_line

FOR each eligible_location:

CALCULATE adjusted_volume

IF adjusted_volume fits:

ASSIGN location

ELSE:

SPLIT in full parcels

CONTINUE search

IF no location found:

SEND to REJET

KEEP status 03

◆ **SECTION 8 — Status Management Matrix**

Condition	Status	Mission Generated
Valid location found	Standard	Yes
Location type = REJET	03	No
Manual reset via menu	01	Relaunch possible

◆ **SECTION 9 — Why This Spec Is Structurally Important**

This document introduces:

- **Volume-driven replenishment**
- **Multi-level stock splitting**
- **Threshold control (ART_NUM2)**
- **Activity-based coefficient logic**
- **Entry point filtering (PE482)**
- **Operational ergonomics changes**

- **Explicit status management logic**

This is now the core ZPA redesign reference document in your knowledge base.